Anqi Fu

Curriculum Vitae

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Academic Employment

2026–Present **NYU Grossman School of Medicine**, New York, NY
Research Assistant Professor, *Department of Radiation Oncology*.

2022–2026 **Memorial Sloan Kettering Cancer Center**, New York, NY
Assistant Attending Computer Scientist (Faculty), *Department of Medical Physics*.

2021–2022 **Memorial Sloan Kettering Cancer Center**, New York, NY
Postdoctoral Research Scholar, *Department of Medical Physics*.

2016 **Stanford School of Medicine**, Stanford, CA
Life Science Research Professional, *Department of Radiation Oncology (Medical Physics Division)*.

Education

2016–2021 **Ph.D. in Electrical Engineering**, *Stanford University*, Stanford, CA
Dissertation: *Convex Optimization Methods for Adaptive Radiation Therapy*
Committee: Stephen P. Boyd (Ph.D. Adviser), Lei Xing, Balasubramanian Narasimhan, John C. Duchi

2012–2014 **M.S. in Statistics**, *Stanford University*, Stanford, CA

2009–2012 **M.A. in Business Research (Economic Analysis and Policy)**, *Stanford Graduate School of Business*, Stanford, CA

2005–2009 **B.S. in Electrical Engineering**, *University of Maryland, College Park*, College Park, MD
Summa cum laude; Electrical and Computer Engineering Chair's Award.

2005–2009 **B.A. in Economics with Minor in Mathematics**, *University of Maryland, College Park*, College Park, MD
Summa cum laude; Dean's scholar; Senior Dillard Award for best undergraduate thesis.

Research Interests

- Radiation Treatment Planning
- Adaptive Radiotherapy
- Radiomic Analysis
- Optimization Algorithms
- Statistical / Machine Learning
- Artificial Intelligence

Works in Progress

Papers

2025 A. Fu, Z. Gouw, J. Jeong, and J. O. Deasy. Optimal radiotherapy dose scheduling with variable fraction sizes and breaks via sequential mixed-integer convex programming. *Physics and Imaging in Radiation Oncology*. Under revision, November 2025.

2024 E. Salanon, A. Fu, A. P. Apte, U. Mahmoud, Z. Belkhatir, A. Shukla-Dave, and J. O. Deasy. Cancer radiomic feature variations due to reconstruction kernel choice and integral tube current. *biorXiv*, June 2024. doi:10.1101/2024.06.04.596806.

Grants

2025 U. Mahmoud, A. P. Apte, A. Dave, A. Fu, A. Plodkowski, J. O. Deasy. Human-informed radiological 3D printed phantoms for harmonizing radiomics and quality assurance of AI tools in cancer imaging. *Imaging and Radiation Sciences Program (IMRAS)*. Letter of intent accepted, November 2025.

2024 A. Fu, U. Mahmoud (co-PIs); S. Hellman (co-I); A. P. Apte, A. Dave, B. Narasimhan, J. O. Deasy (OSCs). Statistical modeling and control of batch effects in radiomics. *NIH/NIBIB R21 Trailblazer Award*. Under revision, December 2024.

Publications

2026 A. Fu, Z. Gouw, J. Jeong, and J. O. Deasy. Optimal radiotherapy dose scheduling with variable fraction sizes and breaks via sequential mixed-integer convex programming. *Physics and Imaging in Radiation Oncology*, 39:101003, May 2026. doi:10.1016/j.phro.2026.101003.

2024 S. Mair, A. Fu, and J. Sjölund. Efficient radiation treatment planning based on voxel importance. *Physics in Medicine and Biology*, 69(16):165031, August 2024. doi:10.1088/1361-6560/ad68bd.

2024 Z. A. R. Gouw, J. Jeong, A. Rimner, N. Y. Lee, A. Jackson, A. Fu, J-J. Sonke, and J. O. Deasy. "Primer shot" fractionation with an early treatment break is theoretically superior to consecutive weekday fractionation schemes for early-stage non-small cell lung cancer. *Radiotherapy and Oncology*, 190(1):110006, January 2024. doi:10.1016/j.radonc.2023.110006.

2024 A. Fu, V. T. Taasti, and M. Zarepisheh. Simultaneous reduction of number of spots and energy layers in intensity modulated proton therapy for rapid spot scanning delivery. *Medical Physics*, 51(8):5722–5737, August 2024. doi:10.1002/mp.17070.

2023 A. Fu, V. T. Taasti, and M. Zarepisheh. Distributed and scalable optimization for robust proton treatment planning. *Medical Physics*, 50(1):633–642, January 2023. doi:10.1002/mp.15897.

2022 A. Fu, L. Xing, and S. Boyd. Operator splitting for adaptive radiation therapy with nonlinear health constraints. *Optimization Methods and Software*, 37(6):2300–2323, June 2022. doi:10.1080/10556788.2022.2078824.

2021 A. Fu, S. Boyd, L. Xing, B. Narasimhan, and J. Duchi. *Convex Optimization Methods for Adaptive Radiation Therapy*. PhD thesis, Stanford University, June 2021. `http://purl.stanford.edu/yk503fd5318`.

2020 A. Fu, J. Zhang, and S. Boyd. Anderson accelerated Douglas-Rachford splitting. *SIAM Journal on Scientific Computing*, 42(6):A3560–A3583, November 2020. doi:10.1137/19M1290097.

2020 A. Fu, B. Narasimhan, and S. Boyd. CVXR: An R package for disciplined convex optimization. *Journal of Statistical Software*, 94(14):1–34, September 2020. doi:10.18637/jss.v094.i14.

2019 A. Fu, B. Ungun, L. Xing, and S. Boyd. A convex optimization approach to radiation treatment planning with dose constraints. *Optimization and Engineering*, 20(1):277–300, March 2019. doi:10.1007/s11081-018-9409-2.

2009 O.J. Glembocki, R.W. Rendell, D.A. Alexon, S.M. Prokes, A. Fu, and M.A. Mastro. Dielectric-substrate-induced surface-enhanced Raman scattering. *Physical Review B*, 80(8):085416, August 2009. doi:10.1103/PhysRevB.80.085416.

2006 R.D. Shull, V. Provenzano, A.J. Shapiro, A. Fu, M.W. Lufaso, J. Karapetrova, G. Kletetschka, and V. Mikula. The effect of small metal additions (Co, Cu, Ga, Mn, Al, Bi, Sn) on the magnetocaloric properties of the $Gd_5Ge_2Si_2$ alloy. *Journal of Applied Physics*, 99(8):08K908, April 2006. doi:10.1063/1.2173632.

2005 J.L. Her, K. Koyama, K. Watanabe, V. Provenzano, A. Fu, A.J. Shapiro, and R.D. Shull. High-magnetic field x-ray diffraction studies on $Gd_5(Ge_{2-x}Fe_x)Si_2$ ($x = 0.05$ and 0.2). *Materials Transactions*, 46(9):2011–2014, September 2005. doi:10.2320/matertrans.46.2011.

Software

General Optimization

A2DR Python solver implementing Anderson accelerated Douglas-Rachford splitting. `github.com/cvxgrp/a2dr`

CVXR R package for disciplined convex optimization. `cvxr.rbind.io`

Biomedical Applications

AdaRad Python library for adaptive radiation therapy with patient health dynamics. `github.com/anqif/adarad`

ConRad Python library for radiation treatment planning with dose-volume constraints. `github.com/bungun/conrad`

OptFrac Python library for optimal dose scheduling in the presence of cellular resource competition, reoxygenation, and hypoxia. `github.com/anqif/opt_frac`

ECHO-VMAT Python library for automated volumetric arc therapy (VMAT) treatment planning using hierarchical optimization. `github.com/PortPy-Project/ECHO-VMAT`

Select Conference Presentations

2024 **INFORMS**, *Seattle, WA*, Optimal dose scheduling with variable fraction sizes and intervals via sequential mixed-integer convex programming

2023 **INFORMS**, *Phoenix, AZ*, Simultaneous reduction of number of spots and energy layers in intensity modulated proton therapy for rapid spot scanning delivery

2022 **AAPM**, *Washington, DC*, Distributed and scalable optimization for robust proton treatment planning

2021 **ECSSC (Invited Workshop)**, *Canberra, Australia*, Convex optimization for statistical and machine learning with CVXR

2021 **EURO**, *Athens, Greece*, Anderson accelerated Douglas-Rachford splitting

2020 **CIRM Conference on Optimization for Machine Learning**, *Marseille, France*, Anderson accelerated Douglas-Rachford splitting

2019 **INFORMS**, *Seattle, WA*, A convex optimization approach to radiation treatment planning with dose constraints

2019 **useR!**, *Toulouse, France*, CVXR: An R Package for Disciplined Convex Optimization

2018 **useR!**, *Brisbane, QLD*, Disciplined Convex Optimization with CVXR

2016 **useR!**, *Stanford, CA*, CVXR: An R Package for Modeling Convex Optimization Problems

Honors and Awards

2018 **Chambers Statistical Software Award**, *Honorable Mention*

2016 **Stanford Graduate Fellowship**, *Stanford University*

2009 **Graduate Research Fellowship**, *National Science Foundation*

2005 **Maryland Distinguished Scholarship**, *University of Maryland, College Park*

2005 **Intel Science Talent Search Semifinalist**

2004 **Siemens Westinghouse Science Competition Semifinalist**

Research Experience

2022–2026 **Assistant Attending Computer Scientist (Faculty)**, *Department of Medical Physics, Memorial Sloan Kettering Cancer Center*, New York, NY
- Designed, developed, and implemented a radiotherapy dose scheduling algorithm that adapts to a dynamic tumor environment and controls for hypoxia-induced radioresistance. Dose schedules are currently being tested in the MSK clinic.
- Analyzed the impact of batch effects on radiomic features. Proposed and tested a new deep learning-based method to harmonize CT scans for radiomic studies.
- Increased the speed and efficiency of ECHO-VMAT treatment planning system by 40% with no loss of plan quality using sparse linear algebra and parallelization.

2021–2022 **Postdoctoral Research Scholar**, *Department of Medical Physics, Memorial Sloan Kettering Cancer Center*, New York, NY
Principal Investigator: Masoud Zarepisheh.
Designed and implemented an ADMM-based distributed algorithm for fast, robust, multi-scenario radiation treatment planning. Evaluated algorithm on proton therapy patient data.

2017–2021 **Graduate Research Assistant**, *Information Systems Lab, Stanford University*, Stanford, CA
Principal Investigator: Stephen P. Boyd.
- Developed and implemented Anderson accelerated Douglas-Rachford splitting (A2DR), a fast parallel algorithm for solving prox-affine optimization problems.
- Led the design, development, and testing of CVXR, an R package that provides an object-oriented modeling language for convex optimization.

2016 **Life Science Research Professional**, *Xing Lab, Stanford School of Medicine*, Stanford, CA
Principal Investigator: Lei Xing.
Developed a unified convex optimization approach for radiation oncology treatment planning with dose-volume constraints. Implemented and tested algorithm on clinical data.

2011 **Summer Intern**, *Economics and Social Systems, Yahoo! Research*, Berkeley, CA
Principal Investigator: Michael A. Schwarz.
Constructed a game theoretic model of market competition when firms have access to targeted advertising technology and analyzed its Nash equilibria.

2009 **Research Assistant**, *Department of Economics, University of Maryland, College Park*, College Park, MD
Principal Investigator: Peter Cramton.
Designed and implemented an ascending clock auction mechanism to efficiently allocate AWS spectrum licenses with regional complementarities and set-aside blocks. Awarded best senior thesis of graduating class.

2008 **Student Researcher (MERIT BIEN Program)**, *Photonics Research Lab, University of Maryland, College Park*, College Park, MD
Principal Investigator: Thomas E. Murphy.
Designed, built, and programmed a scanning laser system. Fabricated single mode porous silicon waveguides for optical sensing of biomolecules and characterized their transmission spectra. Awarded best technical report of MERIT program.

2008 **Research Assistant**, *Department of Electrical Engineering, University of Maryland, College Park*, College Park, MD
Principal Investigator: Mehdi K. Khandani.
Designed, developed, and implemented a Bayesian algorithm to detect and filter incoming packets from DDoS flood attacks.

2004 **Student Researcher**, *National Institute of Standards and Technology*, Gaithersburg, MD
Principal Investigators: Robert D. Shull, Virgil Provenzano.
Characterized the magnetocaloric properties of various metal-doped $Gd_5Ge_2Si_2$ using X-ray diffraction, SEM microscopy and SQUID magnetometry.

Industry Experience

2014–2016 **Machine Learning Scientist**, *H2O.ai*, Mountain View, CA
Led the design, implementation, and testing of generalized low rank models (GLRM) on H2O, a Java-based distributed statistical software engine.

2013 **Summer Intern (Math Hacker)**, *H2O.ai*, Mountain View, CA
Initiated and led the development of an R package that allows users to run H2O in R via the REST API. Package was accepted by CRAN in June 2014.

2008 **Engineering Technician**, *Naval Research Laboratory*, Washington, DC
Collected and analyzed Raman spectra from dielectric core nanowires. Determined optimal Ag deposition time on retroreflector beads coated with SERS-active nanoparticles for greatest signal intensity.

Teaching Experience

Summer 2020 **EE364A: Convex Optimization I**, *Stanford University*, Teaching Fellow
Principal instructor for a class of 100 students from a diverse range of backgrounds. Delivered lectures, designed exams, led discussions of problem sets, and supervised a team of 6 course assistants/graders.

Winter 2020 **EE364A: Convex Optimization I**, *Stanford University*, Course Assistant

Spring 2012 **OIT268: Making Data Relevant**, *Stanford Graduate School of Business*, Course Assistant

Autumn 2007–2008 **ENEE114: Programming Concepts for Engineers**, *University of Maryland, College Park*, Undergraduate Teaching Fellow

References

- **Joseph O. Deasy**
  Chair, Department of Medical Physics
  Memorial Sloan Kettering Cancer Center
  321 East 61st Street
  New York, NY 10065-8204
  Phone: (212) 639-8413
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- **Lei Xing**
  Professor, Department of Radiation Oncology
  Director, Medical Physics Division
  Stanford University School of Medicine
  875 Blake Wilbur Drive
  Stanford, CA 94305-5847
  Phone: (650) 498-7896
  E-mail: lei@stanford.edu

- **Stephen P. Boyd**
  Professor, Department of Electrical Engineering
  Stanford University
  David Packard Building
  350 Jane Stanford Way
  Stanford, CA 94305-9505
  Phone: (650) 723-0002
  E-mail: boyd@stanford.edu